ON ADULT LEARNING

Convert consumption *into capability.*

We watch, we speak, we show up — and most of it doesn't last. The science of why, and the small habits that change it.

Swipe →

Adult professional learning *is mostly consumption.*

Webinars. Conferences. Articles. Podcasts. Panels. Peer conversations. Most adult learning happens in formats designed for delivery — not for retention.

THE FORGETTING CURVE

~70% forgotten in 24 hours.

~90% forgotten in a week.

Ebbinghaus (1885) — replicated repeatedly across 140 years of research.

True regardless of how good the content was, or how engaged you felt at the time.

The most expensive trap *in professional life.*

Cognitive psychologists call it the fluency illusion: the gap between recognising an idea and being able to use it.

WHAT IT FEELS LIKE

Familiarity

You recognise the idea. You can nod along. You feel informed.

WHAT IT IS

Recognition, not knowledge

You can't reproduce it, apply it, or teach it. The change hasn't taken hold.

Adults consume vast amounts of conceptual content. The illusion is feeling well-read while remaining unchanged.

Bjork, Dunlosky & Kornell — Self-Regulated Learning, Annual Review of Psychology (2013)

Modern AI has spent decades *engineering what humans do naturally.*

Three problems machine learning researchers have had to solve explicitly — problems our brains are biologically built to handle, until we stop maintaining the conditions that let them work.

01

IN MACHINE LEARNING

Catastrophic forgetting

Old knowledge gets overwritten by new.

IN HUMANS

Memory consolidation

Sleep, retrieval, and re-encounter protect it.

02

IN MACHINE LEARNING

Transfer learning

Reusing prior models for related tasks.

IN HUMANS

Analogical reasoning

Doesn't happen passively — needs effort.

03

IN MACHINE LEARNING

Stability-plasticity dilemma

Trade-off: holding vs. updating knowledge.

IN HUMANS

Depth vs. currency

The defining tension of professional life.

McCloskey & Cohen (1989); Parisi et al. (2019), Continual Lifelong Learning with Neural Networks

Three principles that travel *from the lab to working life.*

The mechanisms of memory don't care whether you're in a classroom or a Zoom call. What changes is application.

01

Re-encounter beats single exposure.

A second meaningful encounter with an idea — even days later — produces dramatically stronger retention than one intense exposure.

Spacing effect — Cepeda et al. (2006); Latimier, Peyre & Ramus (2021)

02

Retrieval beats review.

Pulling information out of your head — recalling, explaining, applying — strengthens memory more than re-reading or re-watching ever does.

Roediger & Karpicke (2006); Hattie & Donoghue meta-analysis (2021)

03

Effort under difficulty is the signal.

If learning feels easy, very little is being encoded. The struggle to apply or recall is when durable memory is built.

Bjork & Bjork — desirable difficulties (1994, 2020)

A four-stage system *that converts intake into change.*

Principles only matter if you can act on them. What follows is a four-stage system that turns the science into practice — most adults treat learning as a single moment, but it is actually a chain. Break any link and the whole thing collapses.

01 CAPTURE → What you record in the moment.

02 RETURN → What brings you back to it.

03 APPLY → What turns it into behaviour.

04 CALIBRATE — What depth each input deserves.

The next two slides show the moves at each stage.

Notes you can use.
Returns you can't skip.

STAGE 01

Capture well.

- **Paraphrase during input, not after.**
  Generation builds memory; transcription does not. By the time you summarise later, the encoding window has closed.
- **Capture the friction, not the content.**
  Note what the idea contradicts, replaces, or complicates in your thinking. The tension is what your brain holds onto.
- **Write the question, not the answer.**
  Open loops survive; closed ones fade. "Why does this work when X usually doesn't?" outperforms a tidy summary.

Generation effect — Slamecka & Graf (1978)

STAGE 02

Make return inevitable.

- **Return as a question, not a re-read.**
  Close the note. Then ask yourself what it said. The retrieval effort is what builds memory; re-reading produces familiarity, not recall.
- **Build a trigger, not a calendar.**
  The strongest return system surfaces notes when you're tackling a related problem — not on Tuesday at 9am.
- **Tag by the problem it answers.**
  Notes are only retrievable if their language matches the future you who's looking. Tag the problem, not the topic.

Testing effect — Roediger & Karpicke (2006)

The act of capturing creates a false sense of having understood. Without return, the note is the only thing that lasted.

Bridge to behaviour.
Then calibrate the input.

STAGE 03
Apply with intent.

- **Anchor it to something you already do.**
  New behaviours stick when attached to existing routines — the meeting you already run, the report you already write. Empty slots rarely survive.
- **Make it small enough to be embarrassing.**
  A two-minute version that survives beats a thirty-minute version that you skip. Adults consistently overestimate how much friction they'll tolerate.
- **Find the cue, not the willpower.**
  Most application failures are environmental, not motivational. Ask what in your week will force the new thinking to surface.

Implementation intentions — Gollwitzer (1999)

STAGE 04
Match depth to input.

- **Treat unprocessed inputs as a queue.**
  A growing reading list is not learning — it is an inventory of intentions. The score that matters is what's been finished, not what's been started.
- **Decide depth at the start, not the end.**
  Before you press play: is this a passing input, or one that deserves return? "I'll see how it goes" is how most learning leaks.
- **If you can't remember the source, you didn't learn it.**
  Holding an idea without remembering where it came from is the cleanest signal of shallow consumption. Sources keep ideas traceable, challengeable, and yours.

Cognitive load theory — Sweller (1988)

Consume as much as you can digest. The skill is calibration — knowing what deserves a moment of reflection, and what deserves an afternoon.

ONE OBSERVATION

Consumption is what we did.

Capability is what we kept.

The difference is rarely the content. It is almost always what happens after — whether anything was captured, returned to, applied, and given the depth it deserved.